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Relationship between knowledge management processes and performance: critical role of knowledge utilization in organizations

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ABSTRACT

This study examines the relationship between knowledge management processes and its impact on knowledge management performance. A structural model of knowledge management processes showing the relationship of knowledge generation, knowledge sharing, knowledge storage, and knowledge utilization and its subsequent impact on organizational KM performance is proposed. The model is tested using data collected from 1068 individuals working in primarily service based organizations in Turkey. A split-sample methodology is used to validate our measurement model and to further test the structural model. The results indicate that knowledge utilization mediates the relationship between rest of the knowledge management processes and organizational knowledge management performance. Knowledge generation is found to directly impact knowledge management performance, in addition to having a mediated impact through knowledge utilization. The results also provide support for the hypothesized relationship that suggests that the impact of knowledge generation and knowledge sharing on knowledge utilization is mediated by knowledge capture.

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Knowledge management processes; knowledge utilization; organizational knowledge management performance; spit sample; structural equation modelling

1. Introduction

As the world becomes increasingly globalized and organizations face higher levels of competitiveness, what they know and how they manage this knowledge becomes critical for their survival. Though it is clear that organizations engage in various activities in managing their knowledge, it is unclear about the exact mechanism by which they contribute to the organization's performance or how these knowledge management (KM) processes might be related in impacting KM performance. It is also clear that organizations today must evolve into a "knowledge creating company" and must adopt such a view as their organizational culture to stay competitive. (Choi & Lee, 2002; Erden, von Krogh, & Nonaka, 2008; Grant, 1996; Lam, 2000; Lawson & Lorenz, 1999; Nonaka & Toyama, 2003; Zaim & Tarim, 2014). However, there is little empirical evidence regarding how knowledge creation or knowledge generation (in this article knowledge creation and knowledge generation is used interchangeably) is related to other knowledge management processes in contributing to the eventual success of the organization.

KM research in general have maintained two major focus areas in the past. The first stream focuses on knowledge itself and delves into exploring and explaining the knowledge creation and conversion process in organizations. This view often conceptualizes knowledge as tacit and explicit, and focuses on its creation process (Nonaka, 1994). From this perspective, explicit knowledge can be found in written documents and databases in organizations, which can be stored and shared relatively easily. Tacit knowledge, on the other hand, is not readily evident and is mostly retained in individuals' minds, and may consists of their memories and experiences, or may be embedded within the social fabric of the organization (Brown & Duguid, 2001). Knowledge creation, from this perspective, occurs as a result of interplay of tacit knowledge and explicit knowledge in a continual spiral where new knowledge emerges in the context of the specific situations (Bennett III, 1998; Nonaka, 1994; Nonaka & von Krogh, 2009; Polanyi, 1966).

The other stream in KM takes an approach that is more focused on investigating activities involved in creating and managing this knowledge in organizations such as knowledge generation, capture, sharing and utilization in order to gain competitive advantage (Alavi & Leidner, 2001; Martelo-Landroguez & Cepeda-Carrión, 2016). The process of creating knowledge continues with the verification and development of knowledge. From this perspective, knowledge is meaningful when it is codified, classified, given a shape, put in a useful format and stored. Only then, it can be used by the right person, at the right time, in the right way (Alavi & Leidner, 2001; Choi & Lee, 2002; D'Eredita & Barreto, 2006; Kogut

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& Zander, 1993; Nonaka & Toyama, 2003; Schacter, 1981; Zaim, 2006)

Current paper uses this later perspective to identify the various KM processes, and takes a variance based theoretical perspective utilizing a cross-sectional research design to test the relationship between them (Markus & Robey, 1988). In the subsequent sections, we present the relevant literature on knowledge management processes and KM performance. Then we present our conceptual model that capture the relationship between the knowledge management processes and its impact on KM performance. Subsequently, we discuss the overall research method employed for this research. We then present the results of the data analysis of both the measurement model and the structural model performed using structural equation modelling (SEM). Finally, we conclude with the discussion of the findings in the results section acknowledging some of the limitations of this study and presenting its implications.

2. Literature review

In the last two decades KM has progressed from an emergent concept to an increasingly common function in business organizations (Martelo-Landroguez & Cepeda-Carrión, 2016; Zack, McKeen, & Singh, 2009). The literature reveals a rapidly increasing number of studies regarding various aspects of KM and its implementation (Ho, 2009; McAdam & McCreedy, 1999). The importance of KM for organizations' competitive edge seems to be a widely accepted issue in most of the current management literature, acknowledging that effective and efficient management of knowledge leads to superior organizational performance (Kalling, 2003; Magnier-Watanabe & Benton, 2017; Palacios Marqués & José Garrigós Simón, 2006; Schiuma, 2012; Zaim, 2016).

The impact of KM on performance is of considerable interest for KM researchers (Jayasingam, Ansari, Ramayah, & Jantan, 2013). KM performance refers the degree to which KM activities incorporate organizational resources towards achieving KM initiatives (Lee & Choi, 2003). According to knowledge based view, KM initiatives should enhance organization's capabilities to leverage knowledge resources in order to achieve organizational objectives (Grant, 1996). Thus, KM performance focuses on implementation of effective KM, and is distinct from, but connected to organizational performance. The level of KM implementation, and level of individual and organizational efforts in implementing KM initiatives are considered as part of KM performance from this perspective (De Gooijer, 2000; Zaim, Tatoglu, & Zaim, 2007)

Various studies have been carried out in different sectors, countries and cultural contexts with a variety of KM focus areas (Theriou, Aggelidis, & Theriou, 2014). In a recent work, Kianto, Hussinki, and Vanhala (2017), empirically demonstrated how KM practices influence market performance of companies. Their results highlight the key role that knowledge-friendly managers at the helm of the company have in the company's KM journey, and point out knowledge-based learning and developmental activities as effective means to improve market performance. Garcia (2017), worked on groups from Singapore, Indonesia, and the United States, and found empirical evidence that KM affects three basic indicators of organizational performance (sales growth, customer satisfaction, and profit growth) through soft TQM and hard TQM, even though, direct effect of KM on organizational performance was not found. Results of this research suggest that business organizations must promote and execute more KM and soft TQM initiatives in order to influence hard TQM practices, which in turn will positively influence the performance of the organization in the short and long run.

According to Garrick and Chan (2017), the nature of any transfer of tacit knowledge is problematic, even though such employee know-how remains critical to organizational performance and for validating the use-value of knowledge in KM. In another study, Soto-Acosta (2016), indicated that organizations' survival and success depend on the effort and interactions of employees as they carry the skills and generate knowledge to transform new ideas into innovations. They investigated how social web technologies enhance knowledge sharing and collaboration. Specifically, the factors that affect social web knowledge sharing and its effect on innovation performance in manufacturing SMEs were looked at. Abu Naser et al. (2016) examined the relationship of KM maturity and performance excellence in two Palestinian Universities. They found that the most important factors affecting performance excellence were processes, KM leadership, people & KM outcomes.

KM processes have been widely investigated in the literature with very broad and often overlapping classifications. For example, Davenport and Prusak (1998) proposed a theoretical model of KM processes in terms of generation, codification, coordination, and sharing. Alavi and Leidner (2001) indicated that KM processes involved knowledge creation, knowledge storage/retrieval, and knowledge transfer and knowledge application. Similarly, Gold, Malhotra, and Segars (2001) used knowledge acquisition, conversion, protection, and application as part of the knowledge process capability in evaluating its impact on organizational effectiveness. Zaim et al. (2007) provided empirical evidences regarding the impacts of generation, storage, sharing and utilization of knowledge for KM performance. Apart from these few studies, empirical

evidence for the direct and indirect relationships of KM processes and their effect on KM performance has been at best inconclusive. To investigate the possible interplay between the KM processes and its impact on organizational KM performance, this study conceptualizes KM based on the four main processes comprising of knowledge generation, knowledge capture, knowledge sharing and knowledge utilization.

3. Research model and hypotheses

This research aims to explore the relationship between KM processes and the impact of KM processes on KM performance using a variance based theoretical approach as opposed to a process based theoretical approach (Markus & Robey, 1988). While vast majority of the theoretical treatment of KM uses a process based theoretical approach, possibly due to its inherent characteristic and emergent nature, it is not uncommon for empirical studies in KM to use a variance based theoretical approach as is evident in a survey related to knowledge management systems (Matayong & Kamil Mahmood, 2013). From a process based theoretical perspective, KM processes interact with each other and may be difficult to discern a particular direction of relationship between the KM processes. For example, from this perspective, knowledge creation, sharing, and utilization may be viewed as an emergent process where each action contributes to each other in a continual cycle rather than one affecting the other in a dominant relationship (Alavi & Leidner, 2001; Grover & Davenport, 2001). From this perspective they are also sometimes viewed as part of a knowledge process wheel (Carlucci, Marr, & Schiuma, 2004). Another classic example of such an emergent process is Nonaka's (1994) SECI paradigm where knowledge is continuously created in a continual spiral of socialization, externalization, combination, and internalization. From such a process theory perspective it is difficult to establish that one contribute to the other without considering the reverse relationship. A qualitative research design or a longitudinal design will be appropriate in understanding such phenomena from a process perspective. However, the same processes may be studied from a variance theory perspective to understand how the degree of involvement in these processes may impact other outcome variables. For example, in the case of Nonaka's SECI processes, researchers have investigated the impact of these individual processes on KM effectiveness (Sabherwal & Becerra-Fernandez, 2003), or evaluated the four processes as a second order construct (Nonaka, Byosiere, Borucki, & Konno, 1994).

This research conceptualizes KM in terms of knowledge generation, capture, sharing and utilization as the key processes within organizations from a variance theory perspective, and posits that they impact organizational KM performance positively (Jayasingam et al., 2013; Martelo-Landroguez & Cepeda-Carrión, 2016). However, KM processes themselves are not unrelated constructs and may be expected to have some influence within these processes. According to the knowledge based theory of the firm, knowledge creation is a primary reason for the existence of the firms (Conner, 1991; Grant, 1996; Kogut & Zander, 1992). Knowledge creation is also a fundamental process by which organizations try to be competitive in today's knowledge economy (Muhammed, Doll, & Deng, 2011; Nonaka, 1994; Yang & Chen, 2007). Similarly, social capital theory suggest that organizations develop social capital through the interaction of its members and their knowledge sharing (Brown & Duguid, 2001; Putnam, 1993). Hence we posit that these two processes maintain primacy among the four KM processes. Further, before knowledge can be captured or utilized for a particular purpose, it should be created or accessed from other organizational actors. It is also clear from prior research that organizational competitiveness comes from being able to eventually apply or utilize the knowledge they have (Alavi & Leidner, 2001; Davenport & Prusak, 1998). Subsequently we explore the hypothesised relationships in the following paragraphs and present the proposed model in Figure 1.

Knowledge generation and sharing are key activities by which new knowledge is created in organizations (Bock, Zmud, Kim, & Lee, 2005; Muhammed et al., 2011; Nonaka, 1994; Nonaka & Takeuchi, 1995; Von Krogh, 1998; Yang & Chen, 2007). Individuals who engage in these activities can be expected to be motivated to retain this information (Hypotheses: H1a & H2a). They do this by capturing or storing this knowledge either in their own memories or more often in the organizational artefacts, processes, and computer systems (Bollinger & Smith, 2001; Cook & Brown, 1999; Nonaka, 1994). Capturing this knowledge in the organizational documents and systems (or in their own memories) enhances the value of it by making it retrievable for later use in addition for it to being useful at the time it was created or shared. Consequently, knowledge generation, capture, and sharing should lead individuals (and organizations) to be able to use this knowledge and utilize it in some way in their work context within the organization (Hypotheses: H1b, H2b & H3) (Grant, 1996; Koskinen, 2003; Miles, Miles, Perrone, & Edvinsson, 1998; Nonaka, 1994; Nonaka & Teece, 2001).

These actions should also lead to better KM related performance within the organization (Chen & Chen, 2006; Darroch, 2005; De Gooijer, 2000; Jennex & Olfman, 2005; Lee & Choi, 2003; Lee, Gon Kim, & Kim, 2012). However, capturing knowledge or storing

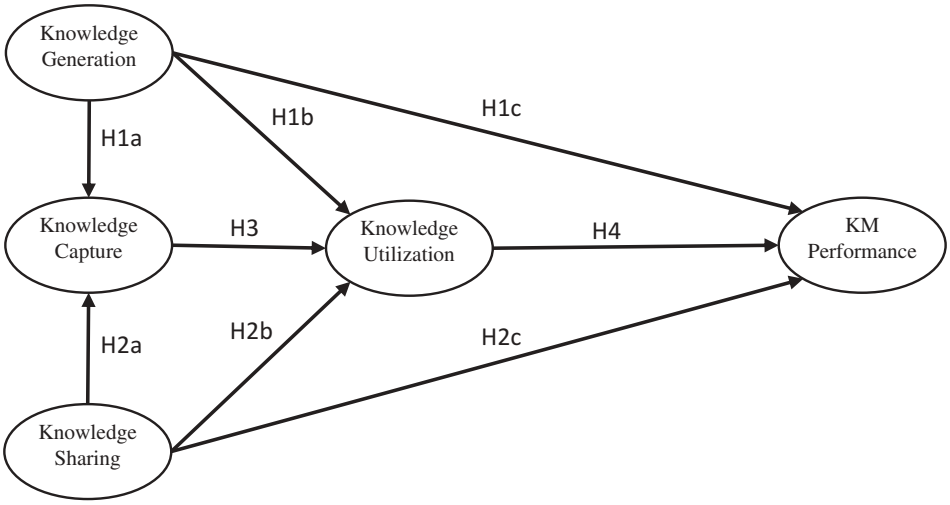


Figure 1. Proposed structural model.

information by itself may not lead to any organizational capability unless it is in a useful form and can be acted upon when necessary (Grant, 1996; Koskinen, 2003; Nonaka & Teece, 2001). Hence, knowledge capture is hypothesized to impact KM performance through knowledge utilization. Thus, capturing knowledge by itself does not contribute value to organizations unless they are acted upon (Hypothesis: H3). In fact, in many cases indiscriminate hoarding of knowledge may lead to information overload and "knowledge clutter" in organizations that can hamper productivity (Edmunds & Morris, 2000; O'Reilly, 1980; Speier, Valacich, & Vessey, 1999). Hence we propose that the link between capture and KM performance is fully mediated by knowledge utilization.

However, knowledge generation and sharing are viewed as more primary processes among the KM processes as discussed earlier, and they are expected to impact KM performance directly (Hypotheses: H1c & H2c), in addition to contributing to it by way of utilizing knowledge (Hypothesis: H4) (Jennex & Olfman, 2005; Kulkarni, Ravindran, & Freeze, 2006; Nonaka, 1994; Nonaka & Takeuchi, 1995). Alavi and Leidner (2001) rightly identify that "the processes of knowledge creation, storage/retrieval, and transfer do not necessarily lead to enhanced organizational performance; effective knowledge application does" (p. 129). Hence, we hypothesize a partial mediation of knowledge utilization on the effect of knowledge generation and knowledge sharing on KM performance due to the potential for these processes to impact KM performance directly in addition to its impact through knowledge utilization.

4. Research method

We used survey methodology to collect data for this study using a questionnaire. Once the data was collected, it was screened for missing data and other data entry errors. In addition to an initial screening of items by a panel of experts before the administration of the questionnaire, we used a two-step split sample approach to assess the measurement model, and for the subsequent test of the hypothesized conceptual model (Anderson & Gerbing, 1988). Initial testing and refinement of the measurement model was done by using the first half of the data that was randomly split into two parts. In the first phase, the measurement model was refined based on factor analysis and content analysis of the items to ensure unidimensionality and validity of the constructs using the first sample. In the second phase, the holdout sample was used to validate the refined measurement model and to test the structural model using structural equation modelling (SEM) in LISREL. SEM provides the ability to model constructs with multiple measurement items and test the relationship between those constructs simultaneously (Jöreskog & Sörbom, 1996). A maximum-likelihood estimation method was used to assess the fit between theorized model and the data. A split sample analysis with a two-step approach is recommended for better validation of the structural model (Anderson & Gerbing, 1988; DeVellis, 2003; Pohlmann, 2004).

4.1. Sample and data collection

Data was collected by distributing 2000 questionnaires to individuals identified as knowledge workers spread across several service based organizations in Turkey. Service sector can be defined as "the portion of the economy that produces intangible goods and services" (Hipp & Grupp, 2005). We received 1068 usable responses yielding a response rate of 53.4 percent.

Knowledge workers are individuals who employ and develops concepts, ideas and theories in their

work, which is often gained through systematic education rather than someone who employs manual skill or physical labour to accomplish tasks (Elliman, Eatock, & Spencer, 2005; Ramírez & Nembhard, 2004). Davenport and Prusak (1998) describes knowledge workers as those who create and use knowledge as a dominant aspect of their work. In selecting the respondents for this study, the HR managers in each of the firms were contacted to identify individuals who met the criteria to be knowledge workers. At a minimum individuals who had a bachelor's degree and who's tasks involved a substantial portion of cognitive work rather than physical work were selected. Further, in order to ensure that individuals were in a position to judge their organization's KM performance only those knowledge workers who were with the organization for at least a year were included.

Most of the respondents were from organizations in finance, education, healthcare, telecommunication, software, and consultancy sectors. Majority of the respondents were also from medium to large organizations that employed 100 or more employees. Medium and large organizations are expected to have a well-established formal or informal KM process. Majority of the respondents in the sample were males (75%). Respondents also represented various levels within the organizational structure, with 7% at the senior management positions and 11% in the mid-level management positions, the rest were non-managerial employees (82%).

4.2. Measures

The survey instrument composed of several questions related to the KM processes and KM performance. The conceptual definition of KM process constructs were adopted from Zaim et al. (2007) and the items were similar to the Organizational Renewal Capability Inventory Survey and the measure used in Kianto, Vanhala, and Heilmann (2016). They had developed a multi-item scale to operationalize the KM processes in services sector, which was adopted to be used in a study investigating the impact of KM processes on leadership (Zaim & Tarim, 2014). The original questionnaire composed of thirty eight items related to the four processes of KM (knowledge generation, capture, sharing and utilization) and KM performance. In the initial measure used for this study, knowledge generation contained nine items, knowledge capture contained eight items, knowledge sharing contained eight items, knowledge utilization contained eight items and KM performance contained five items. Each item was rated on a five-point Likert type scale ranging from strongly disagree (1) to strongly agree (5). The final version of the questionnaire items after item purification and factor analysis are presented in Table A1.

4.3. Common method variance

Bias due to common method variance is a potential problem in studies that use a single source of data. Since the current research used a single informant to collect data on both the dependent and independent variables, several measures were taken to minimise this bias and to assess the extent to which it might affect the substantive relationships tested in this study. Both *ex ante* and *ex post* measures were taken to minimise its effect as suggested by Podsakoff, MacKenzie, Lee, and Podsakoff (2003). As an *ex ante* measure, steps were taken in the design of study's procedures first by paying attention to the scale items to avoid ambiguous and vague terms in the questionnaire, and by keeping the questions simple. Next, respondents' anonymity was assured by stating that clearly in the beginning of the questionnaire. During the data collection process sufficient time was also provided to complete and return the survey thus minimizing evaluation apprehension, all of which have the potential to cause method bias in the data (Podsakoff et al., 2003).

As an *ex post* measure, we evaluated the extent of common method variance by conducting multiple tests. First, Harman's one factor test was conducted by including all items in a principle component factor analysis (Podsakoff & Organ, 1986). Based on the Scree plot and eigenvalue greater than 1, 5 factors were evident explaining a total of 72.41% variance. The first factor accounted for 25.17% of variance and was well below the 50% mark recommended to be deemed problematic for common method variance. Next bivariate correlations between constructs were evaluated, but they were significantly less than one and were not extremely high ($r < 0.90$) thus, suggesting common method variance was not a significant issue for this dataset (Bagozzi, Yi, & Phillips, 1991).

5. Results

The 1068 responses received was randomly split into two samples to facilitate a two phase analysis of the theorized model. The first sample had 533 observations and the second sample (holdout sample) contained 535 observations. Both samples had less than 1% missing data and were distributed randomly throughout the data set and were assessed to be of not an issue for further testing. The first sample with 533 observations was used to test the measurement model, and the holdout sample was used to validate this model and to test the hypothesized structural model.

5.1. Measurement model assessment

The unidimensionality, reliability, and validity of the constructs were assessed using well established procedures (Churchill, 1979; DeVellis, 2003; Nunnally & Bernstein, 1994; Spector, 1992). All items related to the KM processes were factor analysed by forcing the items to the four factors selected in the study. Most items loaded on their respective factors of knowledge generation, knowledge sharing, knowledge capture, and knowledge utilization.

Based on the strength of factor loadings, corrected-item-total correlations (CITC), and any crossloadings that were evident, measures were purified to eliminate multidimensional variables. Item content was evaluated carefully to assess any loss of critical information related to the construct before eliminating them. Eliminating multidimensional variables in favour of unifactorial variables is an acceptable solution if sufficient items remain to operationalize the construct (Anderson & Gerbing, 1988). Unidimensionality of the measurement instruments is a key assumption in measurement theory and is highly recommended for further testing (Gerbing & Anderson, 1988; Hattie, 1985). It simplifies the interpretation of the results as there is less chance of confounding the structural relationships and is an acceptable approach taken by IS and management researchers (Kearns & Lederer, 2003; Segars, 1997; Arend, Zhao, Song, & Im (2017) ; Zhang & Venkatesh, 2017).

In the final measurement model, of the total thirty eight items for all the five constructs combined, only three items that represented the most for each of the construct being measured were maintained for further analysis. All items in the final measurement model loaded on the target factor with at least 0.6 loading and did not have any crossloadings above 0.4 with other factors, indicating unidimensionality.

Next, a measurement model with the final items maintained in the factor analysis was modelled in LISREL to further assess the constructs reliability and validity. Standardized factor loadings and reliability for all constructs are shown in the Table 1. All constructs displayed a sufficient level of reliability as evident from the Cronbach's alpha shown in Table 1, which were above the 0.70 recommended cut-off level. The high factor loadings of items on the target factors that were significant (p-value < 0.05) indicate content validity of the constructs (DeVellis, 2003; Spector, 1992).

All constructs had average variance extracted (AVE) above the recommended 0.5 level indicating good Convergent validity (Fornell & Larcker, 1981). AVE ranged from 0.55 for knowledge generation to 0.72 for KM performance. Several generally accepted goodness-of-fit measures were used to assess the model-data fit (Table 2) (Bentler, 1990; Bollen, 1989; Schreiber, Nora, Stage, Barlow, & King, 2006). The comparative and incremental fit measures indicate a good model-data fit. Absolute fit measures such as chi-square/d.f. are at an acceptable level though slightly above the recommended 2.0.

Correlations of the constructs in the final measurement model is presented in Table 3. All constructs were significantly correlated at p-value < 0.05. Correlations ranged from 0.51 (between knowledge sharing and knowledge capture) to 0.87 (between utilization and KM performance), and were well below the suggested 0.90 level for most items (Bagozzi et al., 1991). To further access the discriminant validity, a chi-square difference test was performed (Anderson & Gerbing, 1988). The results of the test are presented in Table 4. The chi-square statistic for all pairs of unconstrained model was significantly lower than the constrained model. All chi-square differences between the constrained and unconstrained model were significant at p-value < 0.001 indicating discriminant validity of the constructs.

In the second phase of the testing, the holdout sample was used to validate the refined measurement model against the results from the tests conducted with the first sample similar to the approach taken by other researchers in management and education

Table 1. Standardized factor loading from the measurement model, reliability, and AVE (Sample 1).

Construct	Items	Loadings from SEM Measurement Model	Chronbach's Alpha	AVE	Mean	SD
Knowledge Generation	gen2	0.79	0.79	0.55	3.3	0.94
	gen6	0.65				
	gen8	0.78				
Knowledge Capture	store4	0.77	0.86	0.67	3.7	0.93
	store5	0.82				
	store6	0.86				
Knowledge Sharing	share1	0.85	0.83	0.64	4	0.85
	share2	0.80				
	share4	0.75				
Knowledge Utilization	utilize2	0.78	0.86	0.67	3.6	0.88
	utilize3	0.81				
	utilize4	0.87				
Knowledge Management Performance	KMperf3	0.86	0.88	0.72	3.6	0.93
	KMperf4	0.90				
	KMperf5	0.78				

Table 2. Measurement model goodness-of-fit measures (Sample 1).

Goodness of Fit Statistics	Value
Chi-Square	251.82
Degrees of Freedom (df)	80
Chi-Square/df	3.1
Root Mean Square Error of Approximation (RMSEA)	0.064
Root Mean Square Residual (RMR)	0.039
Normed Fit Index (NFI)	0.98
Non-Normed Fit Index (NNFI)	0.98
Comparative Fit Index (CFI)	0.99
Goodness of Fit Index (GFI)	0.94
Adjusted Goodness of Fit Index (AGFI)	0.91

(Ashill & Jobber, 2010; Schreiber et al., 2006). The factor loadings, fit indices and correlations of the measurement model with the holdout sample is shown in the Tables 5, 6, and 7 respectively. The factor loadings, model-data fit and the correlations were similar to the results obtained with the first sample and confirmed the validity of the measures. The confirmatory results from the second sample provided greater confidence in using these measure to further test the structural model. Table A1 lists the final list of items that were retained after the validation of the measures for each construct.

5.2. Structural model assessment

To test the hypothesized relationships and the structural model, data from the second sample (holdout sample) was used to specify the model in LISREL. As this data was not used in specifying the measurement model, it reduces the chance of over specifying the model and bias, and provides greater confidence in the results (Anderson & Gerbing, 1988). The results of the structural model indicate reasonable model-data fit and lend support for most of the hypothesized relationships (Figure 2). Chi-square per degree of freedom is at an acceptable level at 2.7. Other comparative and incremental fit measures indicate good model-data fit as indicated in Figure 2.

The model indicated some error correlations and cross loadings but they were not substantial. Including the error correlations would have provided

Table 3. Correlations (Sample 1, all correlations significant at p-value < 0.01).

	Knowledge Generation	Knowledge Capture	Knowledge Sharing	Knowledge Utilization	Knowledge Management Performance
Knowledge Generation	1.00				
Knowledge Capture	0.62	1.00			
Knowledge Sharing	0.59	0.51	1.00		
Knowledge Utilization	0.75	0.67	0.66	1.00	
Knowledge Management Performance	0.80	0.62	0.58	0.87	1.00

Table 4. chi-square difference tests (Sample 1).

Construct-1	Construct-2	Chi-Sq. Constrained (d.f. = 9)	Chi-Sq. Unconstrained (d.f. = 8)	Difference	Significant at p < 0.001
Knowledge Management Performance	Knowledge Generation	172.44	44.39	128.05	Yes
Knowledge Management Performance	Knowledge Capture	544.19	34.38	509.81	Yes
Knowledge Management Performance	Knowledge Sharing	538.71	27.67	511.04	Yes
Knowledge Management Performance	Knowledge Utilization	162.49	45.43	117.06	Yes
Knowledge Generation	Knowledge Capture	238.98	21.43	217.55	Yes
Knowledge Generation	Knowledge Sharing	220.07	14.55	205.52	Yes
Knowledge Generation	Knowledge Utilization	140.49	24.57	115.92	Yes
Knowledge Capture	Knowledge Sharing	466.63	7.12	459.51	Yes
Knowledge Capture	Knowledge Utilization	283.34	25.95	257.39	Yes
Knowledge Sharing	Knowledge Utilization	363.22	12.54	350.68	Yes

Table 5. Standardized factor loading from the measurement model, reliability, AVE (Sample 2).

Construct	Items	Loadings from SEM Measurement Model	Chronbach's Alpha	AVE	Mean	SD
Knowledge Generation	gen2	0.70	0.75	0.51	3.3	0.94
	gen6	0.64				
	gen8	0.80				
Knowledge Capture	store4	0.79	0.85	0.66	0.37	0.93
	store5	0.84				
	store6	0.81				
Knowledge Sharing	share1	0.80	0.85	0.66	4	0.86
	share2	0.84				
	share4	0.79				
Knowledge Utilization	utilize2	0.75	0.84	0.63	3.6	0.88
	utilize3	0.78				
	utilize4	0.85				
Knowledge Management Performance	KMperf3	0.86	0.88	0.70	3.6	0.94
	KMperf4	0.88				
	KMperf5	0.76				

Table 6. Measurement model goodness-of-fit measures (Sample 2).

Goodness of Fit Statistics	Value
Chi-Square	243.08
Degrees of Freedom (df)	80
Chi-Square/df	3.0
Root Mean Square Error of Approximation (RMSEA)	0.062
Root Mean Square Residual (RMR)	0.039
Normed Fit Index (NFI)	0.98
Non-Normed Fit Index (NNFI)	0.98
Comparative Fit Index (CFI)	0.99
Goodness of Fit Index (GFI)	0.94
Adjusted Goodness of Fit Index (AGFI)	0.91

incremental improvements to the overall model-data fit. Since it is highly recommended not to modify the model solely based on data and is cautioned against adding error correlations without a theoretical rationale, we decided not to include the suggested modifications to improve the fit (Fornell & Larcker, 1981). Further, addition of these suggested modifications even though improved the model fit, they did not have any substantial impact on the structural loadings and the interpretation of the relationships between the constructs. However, one error correlation between the two items (KMperf4 and KMperf5) in the KM performance construct was included in the final model as these error correlations were within the construct and the items contained similar wordings.

All hypothesized structural paths were significant (p-value < 0.05) except for the direct path from knowledge sharing to KM performance (Table 8). However, knowledge sharing had a significant and substantial indirect effect (Table 9) on KM performance suggesting that knowledge utilization fully mediates the relationship between knowledge sharing and KM performance. However, to establish full mediation it is important to first establish a significant path from the predictor variable to the outcome variable in the absence of the mediating variable (Mathieu & Taylor, 2006). Hence the path from knowledge sharing to KM performance was tested directly without knowledge utilization. The results indicated that knowledge sharing had a significant effect on KM performance which became non-significant with the introduction of knowledge utilization suggesting that knowledge utilization fully mediate this relationship. As hypothesized, the relationship between knowledge capture and KM performance is also fully mediated by knowledge utilization. The relationship between knowledge generation and KM performance is partially mediated by knowledge utilization as hypothesized initially, and were tested following the approach suggested by Mathieu and Taylor (2006).

The summary of the results related to the hypothesized relationships based on the structural model is

Table 7. Correlations (Sample 2, all correlations significant at p-value < 0.01).

	Knowledge Generation	Knowledge Capture	Knowledge Sharing	Knowledge Utilization	Knowledge Management Performance
Knowledge Generation	1.00				
Knowledge Capture	0.60	1.00			
Knowledge Sharing	0.65	0.64	1.00		
Knowledge Utilization	0.77	0.73	0.67	1.00	
Knowledge Management Performance	0.75	0.60	0.59	0.86	1.00

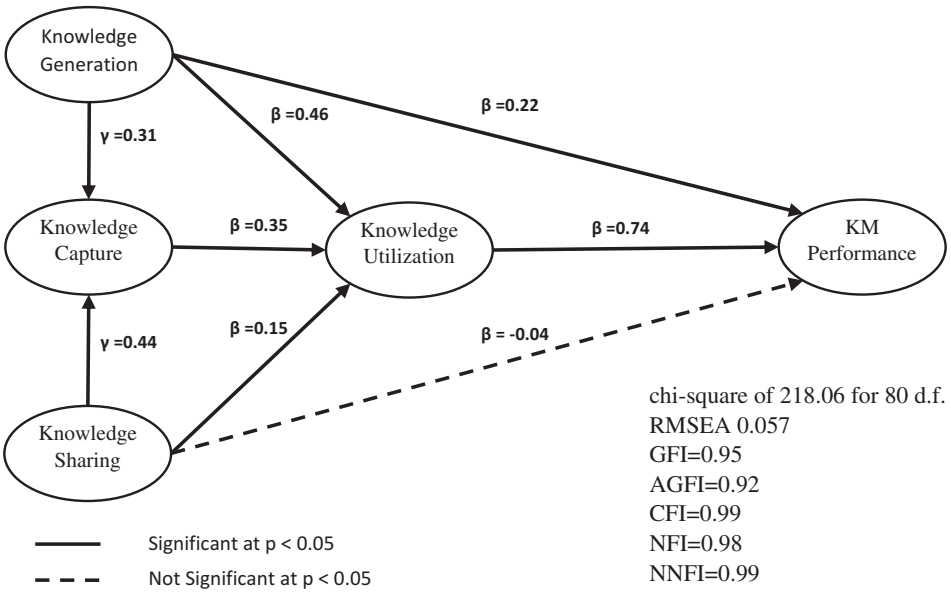


Figure 2. Standardized loadings of the structural model.

Table 8. Summary results of the hypotheses based on the structural model.

Hypotheses	Standardized coefficient	t-value	Supported
H1a: Knowledge generation is positively associated with knowledge capture	0.31	4.79	Yes
H1b: Knowledge generation is positively associated with knowledge utilization	0.46	7.53	Yes
H1c: Knowledge generation is positively associated with knowledge management performance in organizations	0.22	3.15	Yes
H2a: Knowledge sharing is positively associated with knowledge capture	0.44	6.91	Yes
H2b: Knowledge sharing is positively associated with knowledge utilization	0.15	2.46	Yes
H2c: Knowledge sharing is positively associated with knowledge management performance in organizations	−0.04	−0.85	No
H3: Knowledge capture is positively associated with knowledge utilization	0.35	6.32	Yes
H4: Knowledge utilization is positively associated with knowledge management performance in organizations	0.74	9.82	Yes

Table 9. Direct, indirect and total effect of KM process constructs on KM Performance.

Constructs	Direct effect	Indirect effect	Total effect
Knowledge Generation	0.22	0.42	0.64
Knowledge Capture		0.26	0.26
Knowledge Sharing		0.22	0.22
Knowledge Utilization	0.74		0.74

presented in Table 8. The structural model with the significant paths are shown in Figure 2. All paths to knowledge utilization were significant, along with the path from knowledge utilization to knowledge performance providing support for our main contention that knowledge utilization mediates the relationship of other KM processes to KM performance. The total effect of knowledge sharing to knowledge utilization was 0.30 (Beta), and knowledge capture to knowledge utilization was 0.35 (Beta). KM processes combined explained 79% of variation in KM performance based on the R-square. Knowledge generation, sharing, and capture combined explained 71% of variation in knowledge utilization. Knowledge generation and sharing combined explained 47 % of variation in knowledge capture.

6. Discussion

Table 9 presents the summary of direct, indirect and total effects of all significant relationships in the structural model. Data shows that knowledge utilization has the strongest relationship to organizational KM performance. It is consistent with our contention that a key capability in the success of KM is the organizations' ability to utilize the knowledge resources that is available to them. It is also consistent with other organizational theories such as resource based view of the firms (Barney, 2001; Wernerfelt, 1995). Utilization is also cited as one of the key processes within the KM literature (Grant, 1996; Koskinen, 2003; Nonaka, 1994; Nonaka & Teece, 2001).

Results substantiate our central argument that knowledge utilization has a critical role among the KM processes in converting the KM process capability of organizations to KM performance, and is consistent with other theoretical and empirical findings that connect knowledge utilization with organizational performance (Davenport & Prusak, 1998; Lee & Choi, 2003; Zaim, 2016). However the mediating role of knowledge utilization is rarely investigated. For example, Martelo-Landroguez and Cegarra-Navarro (2014) analysed the direct and indirect effects of knowledge application on customer value chain. Another empirical study (Zhang, Di Benedetto, & Hoenig, 2009) revealed the mediating role of knowledge utilization with respect to product development strategy and innovation performance. Results reveal that knowledge utilization fully mediates the relationship of two of the four KM processes (sharing and capture) on KM performance, and partially mediate the relationship of knowledge generation on KM performance. Accordingly, knowledge utilization appears to be the central process through which rest of the KM processes: knowledge generation, knowledge sharing and knowledge capture, impacts the KM performance.

The second largest total effect on KM performance was due to knowledge generation. The KM literature is replete with evidences that indicate knowledge generation/creation is a core capability in organizations in creating value (Davenport & Prusak, 1998; Nonaka & Takeuchi, 1995; Von Krogh, Nonaka, & Aben, 2001). Firms' ability to generate new ideas and knowledge is a critical and non-imitable resource that can provide them competitiveness and as such it is expected to impact their KM performance. Direct and indirect effects of knowledge generation and development on performance, sustainability and competitive advantage have been widely investigated in KM literature (Choi, Poon, & Davis, 2008; Magnier-Watanabe & Senoo, 2009). Our findings confirm this and show that not only does knowledge generation contribute to the success of KM in organization by enabling organizations to use the knowledge that is created, but also the process of knowledge generation by itself contribute value for organizations as is evident by the direct relationship to KM performance after taking into consideration its impact through knowledge utilization.

The third significant finding is that the impact of knowledge generation and knowledge sharing on knowledge utilization is mediated by knowledge storage or capture. Capturing, storing and codifying knowledge is clearly considered as one of the key processes of KM and the complex relationship of knowledge capture with knowledge sharing and utilization, has been extensively analysed (Coakes, 2006; Merali & Davies, 2001). Nevertheless, the mediating effect of knowledge capture with respect to KM processes has relatively attracted less attention. Jasimuddin and Zhang (2011) analysed the relationship between knowledge storing and transfer and conclude that these are interactive and dynamic processes and an integrated approach is required for successful knowledge transfer. Choi et al. (2008) also explains the complementarity of KM processes for effective organizational performance. There are many other theoretical research findings exploring the significant role of capturing, storing and codifying knowledge for effective KM implementation (Coakes, 2006; Martelo-Landroguez & Cegarra-Navarro, 2014). However, there is lack of empirical evidences regarding the mediating role of knowledge capture with respect to other KM processes which our findings clarify to some degree.

Knowledge sharing and capture even though, did not have a direct impact on KM performance, they had a significant indirect effect on KM performance. Further, knowledge sharing and knowledge capture remain important KM processes within organizations. Our data suggests that they contribute to KM performance by enabling organizational actors to utilize the knowledge that is being shared and captured within the organizations. The findings are consistent with prior research that found similar relationship where knowledge sharing impacted task knowledge through knowledge application (Muhammed et al., 2011), and had no direct effect on other organizational measures such as organizational value creation (Martelo-Landroguez & Cepeda-Carrión, 2016) and process improvements (Jayasingam et al., 2013).

6.1. Limitations

This research presents strong evidence for the proposed structural model linking KM processes to KM performance, and for the central role of knowledge utilization in this relationship. Results provide additional credibility for the support of the theoretical model due to the split sample methodology that was adopted in this research. The holdout sample provides additional confirmation since it enables an independent model re-estimation. However, the results should be interpreted within the limitations of the study. First, the data collected was limited to only services sector organizations operating in Istanbul, Turkey. Though this severely limits the generalizability of the study, and should be interpreted within the cultural context, it is unlikely that these organizations operate differently in relation to KM processes to organizations in other parts of the world. Further, Turkey being a connecting point between east and west many of the organizations were part of multinational corporations or had presence outside Turkey. In spite of these considerations, further research should be conducted to validate this model, especially in the light of cultural peculiarities such as those that are identified by the GLOBE study (House, Hanges, Javidan, Dorfman, & Gupta, 2004) or that of Hofstede (2003), and in other organizational contexts beyond those of service industries.

Another potential limitation is that the data was collected based on convenience sampling. The fact that questionnaire was administered in Turkish may also be a limiting factor because, it may have the potential for the measurement items to be interpreted differently in the context of this target language. Though care was taken to translate the items as reported in Table A1 so that it can be available for a wider audience, they need to be further tested to reassess the reliability and validity of the measures in future studies.

6.2. Implication for theory

Knowledge management process has been defined in the literature as comprising of a variety of processes that sometimes overlap each other (Alavi & Leidner, 2001; Earl, 2001; Hussinki, Kianto, Vanhala, & Ritala, 2017; Kogut & Zander, 1993; Spender, 1996; Teece, 1998; Theriou et al., 2014). We find that they can be modelled usefully as knowledge generation, knowledge sharing, knowledge capture, and knowledge utilization in an organizational context. The measures used in this study provide scales that are reliable and valid to be used in future KM research. Most KM research treat these processes as independent and contributing to the overall KM process in spite of the theory suggesting relationships between them. This paper explored these relationships and found support for knowledge generation and sharing to be viewed as more elementary processes in organizations as they help build organizations' knowledge capability which they can utilize better for increased performance. This is in line with prior research suggesting the importance of knowledge generation and sharing for organizations (Bock et al., 2005; Muhammed et al., 2011; Nonaka, 1994; Von Krogh, 1998; Yang & Chen, 2007).

Knowledge development theory claims that the link between knowledge and performance is not very explicit or automatic though it is essential to

generate and share knowledge in order to capture and utilize it (Kalling, 2003). Further, knowledge generation and sharing processes are usually considered as the initial processes of KM that leads to knowledge application or utilization (Grover & Davenport, 2001; Mills & Smith, 2011; Zaim et al., 2007). This is also evident in the results of this study. Organizations can achieve KM performance when they create and share knowledge within the organization which they should capture and utilize for an even better KM performance. The results also clarify the central mediating role of knowledge utilization in the impact of KM process on KM performance. This research will help future researchers exploring the outcomes or antecedents of KM processes to consider the relationships between knowledge management processes whereby contributing to a more meaningful interpretation of KM in organizations. The results contribute to the theoretical explanation of KM constructs and their relationships from a variance based perspective and helps connect it to the process theory perspective that is dominant in the KM literature (Markus & Robey, 1988).

6.3. Implication for practice

Organizations are key structures in today's society where new knowledge is created and applied to generate value. This study investigates the relationships between various KM processes and the key role of knowledge utilization in improving KM related performance in organizations. Organizations need to pay special attention to all KM processes such as knowledge generation, sharing, capture and utilization in order for an enhanced KM performance that could eventually lead to competitiveness and improve the performance of the organization itself. Though there has been substantial emphasis regarding knowledge generation/creation as a key capability that successful organizations pursue, the findings suggest that organizations should also investigate ways to encourage its employees to apply the knowledge that they create and retrieve from others for that knowledge to have a greater impact on their KM performance. Managers responsible for implementation of KM in organizations should explore ways to encourage better utilization of individual and organizational knowledge that may be available to their employees. The research has practical implications by enabling mangers to understand how various KM processes occurring within the organization may contribute to KM performance within their organizations.

7. Conclusion

Though there are several studies examining the impact of KM processes on various individual and organizational outcomes and performance measures, our study is the first to explore the relationship between KM processes in the organizational context from a variance theory perspective. KM literature is rich with theoretical treatment regarding the importance of knowledge utilization or application and its connection to various organizational performance measures, however, there are few studies that empirically examine this connection. This article contributes to the KM literature by providing empirical support for this contention and for the central role of knowledge utilization in the relationship of KM processes to KM performance. Though we did not explicitly test the link from KM performance to organizational performance, this is implied in this research. Some evidence for such link can be found in prior literature on KM (Ahmad, Lodhi, Zaman, & Naseem, 2017; Darroch, 2005; Davenport & Prusak, 1998; Gold et al., 2001; Nonaka, 1994). Drawing on a variance based perspective of KM, this research supports that knowledge generation and knowledge sharing are rather more primary processes within KM that drives knowledge capture and knowledge utilization in organizations. For organizations to translate this to a better KM related performance and ultimately to their organizational performance itself, they should be able to utilize the knowledge that is being created, shared and captured there.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix

Table A1. Final measurement items for knowledge management process constructs and knowledge management performance.

Label	Measurement item	strongly disagree	disagree	undeci-ded	agree	strongly agree
Knowledge Generation						
gen2	all employees are encouraged to learn continuously	1	2	3	4	5
gen6	employees are encouraged to make recommendation	1	2	3	4	5
gen8	employees are involved in knowledge generation processes	1	2	3	4	5
Knowledge Sharing						
share1	we give a special emphasis on knowledge sharing with our colleagues	1	2	3	4	5
share2	e-mails, mobile applications are effectively used to share knowledge among our colleagues	1	2	3	4	5
share3	we have effective communication with our colleagues	1	2	3	4	5
Knowledge Capture						
store4	information related to customers, products and services are effectively captured and stored	1	2	3	4	5
store5	we keep up-to-date (relevant) information about our products, services	1	2	3	4	5
store6	we have up-to-date (relevant) information about our suppliers and competitors	1	2	3	4	5
Knowledge Utilization						
utilize2	we have accurate and effective decision making process	1	2	3	4	5
utilize3	my organization can utilize its knowledge base	1	2	3	4	5
utilize4	we are encouraged to implement what we know in our job-related activities	1	2	3	4	5
Knowledge Management Performance						
KMperf3	I think this organization implements KM successfully	1	2	3	4	5
KMperf4	I think this organization is spending enough time and effort to implement KM successfully	1	2	3	4	5
KMperf5	I think this organization is making the required investments to implement KM successfully	1	2	3	4	5